

Technical specifications to meet the requirements of an Automatic Code Compliance Checking tool and current developments in infrastructure construction

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ABSTRACT

Building Information Modeling has revolutionized the construction industry by, for example, contributing to the automation of different activities related to the design, planning, and construction of civil works. Tasks such as the review of construction-related codes and regulations, which today still require a great deal of human effort, have begun to be automated through the development of digital applications and tools, thus resulting in the BIM use case Automated Code Compliance Checking (ACCC). So far, however, the research and development around this topic have been focused mainly on the building area, leaving a gap for the infrastructure sector. With this article, the authors intend to contribute to closing this gap, by providing an overview of the current state of development of ACCC tools in infrastructure construction. This overview supports the identification of challenges and requirements for the development of such tools, which are also discussed in detail in the current work. Finally, the author's considerations regarding the formalization and structuring of the engineering knowledge and infrastructure data required for the development of ACCC tools, as well as its mapping in a graph database, which allows access to all the necessary and relevant information for the code checking process, are presented. The result of the research carried out is the detailed structuring of the regulations and their rules from infrastructure construction and the possible modeling implementation in a graph database. In practice, this should help the leading software manufacturers in cooperation with infrastructure engineers, to develop a full-fledged ACCC tool.

1. Introduction

Regulations represent the knowledge in the form of rules, that support the achievement of optimal planning and construction of infrastructure, considering aspects such as cost, safety, quality, and uniformity. They are based on scientific findings, knowledge, technology, and practical experience. Thus, they aim to define a certain basis for ensuring a safe, efficient, and profitable planning, construction, and operation, and to make this basis available for engineers. Additionally, in transport infrastructure, regulations play a major role in ride comfort [36,42]. Compliance with regulations must be ensured over the entire lifecycle of a construction work [30,47] not only because it contributes to avoid cost overruns and ensure safety and quality, but also because it is an essential prerequisite for its approval and subsequent implementation and operation [30,36,61].

However, there are still many challenges to overcome related to code

compliance checking in infrastructure construction. An example of this is the complexity of infrastructure planning, which usually involves iterative processes and requires the consideration of many social, political, economic, environmental, and technical variables. Usually, compliance with many different regulations (various degrees of obligation and technical standards, etc.) is required and must be ensured, which means that multiple regulations are needed at the same time (see example in Fig. 1) - either for multiple independent technical circumstances that need to be reviewed within a single project or for reviewing one single circumstance that needs to comply with multiple regulations at the same time.

Additionally, the fact that planning is usually an iterative process leads to the need of performing the regulation-checking process several times [61]. Keeping track of all the required regulations can be very complicated, which may cause some regulations to be overlooked, resulting in the infrastructure not following the regulatory standards.

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Non-compliance can be very problematic, as it can lead to serious errors, which will inevitably be reflected in the time and cost of the project, not to mention the safety of the infrastructure. Moreover, the increasing advances in technological developments in civil engineering (e.g., the development of digital building modeling), have made it possible to construct more complex structures and thus projects, which, in turn, require the constant change and adaptation of regulations, posing a major challenge to designers [8,50,61], who need to keep up with these modifications and ensure that their designs are carried out following the pertinent regulations. Another issue related to code compliance checking is the fact that regulation rules can be complex and abstract in nature with ambiguous interpretability, making the manual review of a project increasingly time-consuming and prone to errors [30,34,50,59,61]. In addition, regulations can be poorly structured, incomplete, contradictory, and may have subjective content [30,36,42]. They contain gradations of content and a multitude of exceptions [42].

Nowadays, most of the regulations that support the design and construction of infrastructure are still in textual form and written in a way that can be interpreted by humans, but not by machines (see example in Fig. 2). Thus, the designs still need to be checked mostly manually, using documents in paper form (2D plans, regulations). All the above makes regulation checking and reviewing increasingly time-consuming and an error-prone process due to human subjectivity and the possibility of not detecting all regulatory violations.

In this context, the BIM use case Automatic Code Compliance Checking (ACCC) has surged as a solution to the aforementioned issues leading to the development of so-called ACCC tools. These tools help to minimize the manual effort of checking processes for compliance with the regulations in a design, thus relieving engineers and designers and allowing to systematically check attributes, parameters of objects, relationships between individual objects, etc, which are required by the necessary regulations in a planned construction. As a result, non-compliant circumstances are displayed. These tools have the potential to make the code checking process more efficient and accurate, compared to conventional checking methods based on the 2.5D planning methodology. In order to maximize the use of all these potential benefits, the processes, applications, and tools related to ACCC must be designed in such a way that ensures the completeness and availability of all relevant data when performing the code checking process. In this article, the authors intend to describe their considerations regarding the information requirements when checking infrastructure projects against codes and regulations as well as the storage and management of this information by means of a database of regulations and regulation rules, which should support the design and development of checking tools (ACCC tools) for the infrastructure sector.

In this paper, the structure and composition of the regulation rules in the regulations for infrastructure construction as well as general

6.1.7.2 Cycling provision on the carriageway

On low-traffic roads and on roads subject to low motor vehicle speed limits (such as 30 km/h zones), cycle traffic can generally be comfortable and travel quite safely on the carriageway. On other roads, measures should be investigated in order to enhance the safety and comfort of carriageway use. This includes:

- enhanced precautions to ensure that the maximum speed limit is observed;
- reducing the permissible maximum speed.

Suitable parameters are carriageway widths up to 6.00 m at low traffic volumes, up to 500 vehicles per hour and carriageway widths of over 7.00 m up to medium traffic volumes from 800 to 1,000 vehicles per hour with 6 % heavy traffic. At a speeds of $V_{85} < 50$ km/h and with little or no heavy traffic, the traffic volumes above may be exceeded in individual cases.

Fig. 2. Example of a text passage from the RAS 06 Directives for the Design of Urban Roads for guiding cycle traffic on the carriageway for motor vehicles [21].

requirements are examined and summarized on the basis of the German regulations for track and road construction, as well as previously published research work. Requirements for machine-readable implementation in a database are also derived. For this purpose, two separate classifications (one for the regulations and one for the regulation rules) are proposed, and the principal model of the interaction between regulations, regulation rules and infrastructure objects in a graph database is shown at a high level of abstraction. In the practice, the identification of requirements and the associated difficulties should help software developers to elaborate concepts for the implementation of an ACCC tool in a next step, considering the software-technical aspects, and then to develop a fully-fledged software for the implementation of the BIM use case of automatic regulatory compliance checking of a planning.

2. Automatic code compliance checking in infrastructure planning

In building construction, several approaches for ACCC have been developed. Among these, there are logic-based (such as propositional and predicate logic or deontic logic, e.g., in the form of decision tables), language-based (such as markup language), and ontological approaches. Usually, these approaches focus rather on making the regulations and

	DIN EN 13201-1	Street lighting; selection of street lighting classes
	DIN EN 13201-2	Street lighting; quality characteristics
	DIN EN 13201-3	Street lighting; calculation of quality characteristics
	DIN EN 13201-4	Street lighting; methods of measuring the quality characteristics of street lighting installations
FGSV ²⁾		Standard vehicles and tractrix curves for checking of the negotiability of traffic areas (FGSV 287)
EAE		Recommendations for the layout of access roads (FGSV 285) (The EAE have been replaced by FGSV 200; RAS 06.)
EAHV		Recommendations for the layout of main arterial roads (FGSV 286) (The EAHV have been replaced by FGSV 200; RAS 06.)
EAR		Recommendations for resting traffic installations (FGSV 283)
EFA		Recommendations for pedestrian traffic installations (FGSV 288)
ERA		Recommendations for cycle traffic installations (FGSV 284)
ESAS		Recommendations for safety auditing of roads (FGSV 298)
ESG		Recommendations for street space design in built-up areas (FGSV 230)
ESN		Recommendations for safety analysis of road networks (FGSV 383)

Fig. 1. Exemplary excerpt from the RAS 06 Directives for the Design of Urban Roads regarding the additional regulations to be observed when planning urban roads [21].

their content interpretable by machines than on the communication with the users (engineers, planners, constructors), thus neglecting human-machine communication [1,6,7,30,31]. In addition, many of the currently available software solutions have limited testing capabilities, allowing only the translation of simple rules, and are, therefore, not suitable for more complex logic [6,7]. For example, solutions such as parametric tables can be an easy way of representing rules, since no programming knowledge is required, but have limited capabilities when it comes to complex rules (restricted range of conditions) [6].

According to Ref. [6], the approaches for developing digital and machine-readable representations of regulation rules can be separated basically into hard-coded and logical, the former being more complex and requiring programming knowledges. The authors of [31] identify the following approaches:

- “Procedural code,
- Sets of rules or decision tables,
- Objects with attributes and methods associated with those attributes,
- Formal predicate logic expressions,
- Description logic approaches,
- Context-oriented approaches,
- Simulation-based approaches,
- Agent-based approaches, and
- Other hybrid approaches“ [31]

In 2015, the study [6] presented an overview of the software solutions for ACCC in building construction. Later, in 2020, the authors of [50] contributed with a detailed summary of different possible rule interpretations, by presenting approaches such as hard-coded, object-based, logical, and ontological-semantic.

From the literature review, it was founded that the research around the topic of automatic code compliance checking is very active. However, most of the publications and contributions are related to building construction and only a few focus on infrastructure. Some of the attempts in the area of ACCC for infrastructure founded in the literature are presented below.

A (semi-)automatic tool for regulation conformity check was developed in 2015 using the software Excel [36]. Here, the authors deal with the elements of alignment both in the vertical and the horizontal planes, as well as cant and, to some extent, with the application of forces on the track bed of a railroad line, according to Brazilian regulations. In the developed ACCC tool, as usual, the input data are requested at the beginning, which are to be fed from the alignment-design tool into the Excel table. Further input information, such as speed and track gauge, is also to be provided. Having this information, the tool starts checking section by section of the infrastructure for minimum/maximum or permissible/inadmissible values.

Efforts to ACCC for the construction of underground utilities (sewage, gas, electricity, water and telecommunications supply lines, etc.) are also done in Ref. [60]. In Ref. [60], an ontological approach is presented, which should allow the automatic verification of the position (verification of spatial rules) of the individual pipelines relative to each other, as well as to other adjacent infrastructures, such as roads and/or rails, both in the vertical and horizontal direction, in the urban area. The approach should be applicable for new as well as for existing utilities.

Other approaches focus on the creation of user-friendly representations of regulation rules in an ACCC tool, that would require less programming knowledge while still allowing the development of a machine-readable representation of complex rules, by means of visual programming languages (VPL). A great advantage of these languages is that they can depict the syntax of regulation rules through graphical elements that can be easily read and understood by both humans and machines [7]. The possibility of using VPL for code compliance checking in the railroad sector was investigated in Ref. [39]. On one hand, the publication evaluated the extent to which the examined guidelines can be automated. On the other hand, the possibility of using the RASE

(Requirement Applicability Selection Exception) method [1,53], the BPMN (Business Process Model Notation) and DMN (Decision Model and Notation) to translate regulation rules into a machine-readable language was investigated. The RASE method is used as a preprocessing step to support the translation of textual information, which is then structured as a workflow diagram using standardized symbols from BPMN and DMN (see Fig. 3).

At each step of the process, the rule is broken down into small components, which ensures better understanding. The formalized and structured regulation-related content should be later used in KBE (Knowledge-Based Engineering) systems [37,39], with the idea of making the technical knowledge from the planning of railroad infrastructure (e.g., alignment, superstructure dimensioning) digitally available for computer processing purposes and then link it to CAD (Computer-aided design) via the KBE.

To test the developed approach, the authors of the present article [39] used the guidelines from the *National Railway Company of Germany* (DB AG, by its German acronym) [10–14] as examples. However, the proposed method is not exclusively applicable to the railroad domain, thus it should be transferable to other fields, such as road-related regulations (cross-section design and superstructure dimensioning), and applicable to other countries as well. A total of 943 rules were evaluated (only those required for design), of which 486 (52%) are automatable and 46% are suitable for being represented using BPMN and DMN notations. It was shown that the degree of automation depends on the deterministic description (explicit information such as height, width, distance, radius, etc. about the infrastructure objects) of the rule. In this sense, 82% of the rules belonging to the *800 family* guidelines of DB AG are parametric, thus the suitability of using BPMN and DMN is 85%. The proportion of parametric rules contained in Refs. [12,13] is 38% and 39%, respectively, which means that the applicability of the BPMN and DMN notations is lower.

Despite the aforementioned contributions made to the field of ACCC for infrastructure, a generally applicable model that provides a uniform and standardized representation of infrastructure-related regulations and rules is still missing [39]. According to Refs. [37,53], currently there are no standardized procedures and methods for the digital and machine-readable representation of engineering knowledge, including regulation rules, in railway construction. The software tools available on the market offer libraries of construction objects, which allow, to a certain extent, the digital representation of these objects and the relationships between them [37]; however, these representations might not be detailed enough to fully support the checking process. In general, there is still a large research gap in the field of infrastructure construction.

3. Specifications for the development of an automatic code compliance checking tool

When translating regulations to a machine-readable language, it is not only important to translate the regulation rules, but also additional information must be considered. This includes, for example, data such as country or regional context of the regulation, type of regulation, and degree of obligation. In this section, the aspects considered relevant for developing an ACCC tool in the context of infrastructure construction are discussed.

The aspects and their description are supported by the reviewed literature [4,5,13,15–21,29–31,39,41,47,52,54,57–59,61] as well as by the author’s considerations about the difficulties that might occur during the development of the tool.

The general aspects of the digital representation of the regulation and regulation rules are primarily considered, because this represents the basis of the ACCC process. Once the aspects are defined, it should be decided at which level each aspect is to be implemented, that is, either at the regulation level or at the regulation rule level. The next step is to think about the regulation and rule modeling process itself.

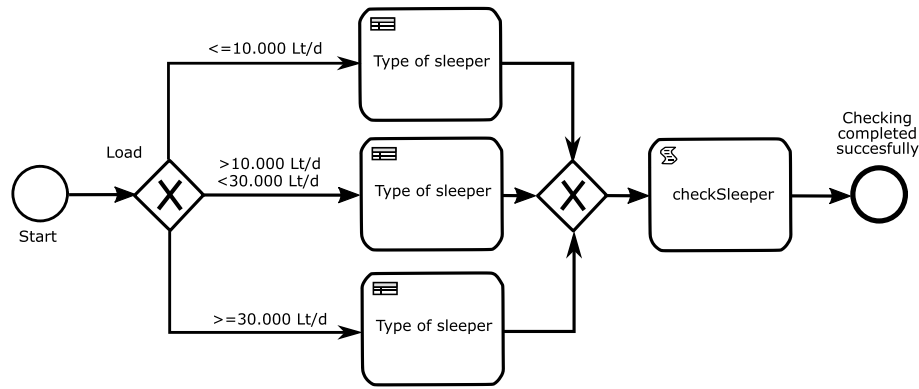


Fig. 3. Representation of the workflow using BPMN and DMN for checking the correct choice of sleepers differentiated by track load ranges according to guideline [12] presented in Ref. [39].

For this, a graph database is considered due to the following reason: The regulations from which the individual dependencies between the infrastructure objects result, can contain different kinds of information of different characters (obligatory or not) at the same time. These pieces of information are usually structured in an irregular, disorganized and complex manner. For example, there might be several references to several technical circumstances that must be considered, both within the same regulation and across several regulations. In the end, this creates a large list of requirements to be fulfilled, which can be confusing and

difficult to follow. For this purpose, graph databases are well suited since they provide a good representation of highly networked or chained and heterogeneous data(sets), especially if the data complexity exceeds the storing capacities of relational databases. Graph databases allow then an easier and more efficient management of complex data and improve visualization for the user [48]. The edges, which are an essential part of a graph database, generally allow a simple representation of the relationships between objects (nodes).

The individual aspects are presented and discussed in detail below,

Name: ACCC tool
 Package: Requirements ACCC tool (englisch)
 Version: 1.0
 Author: ac123807

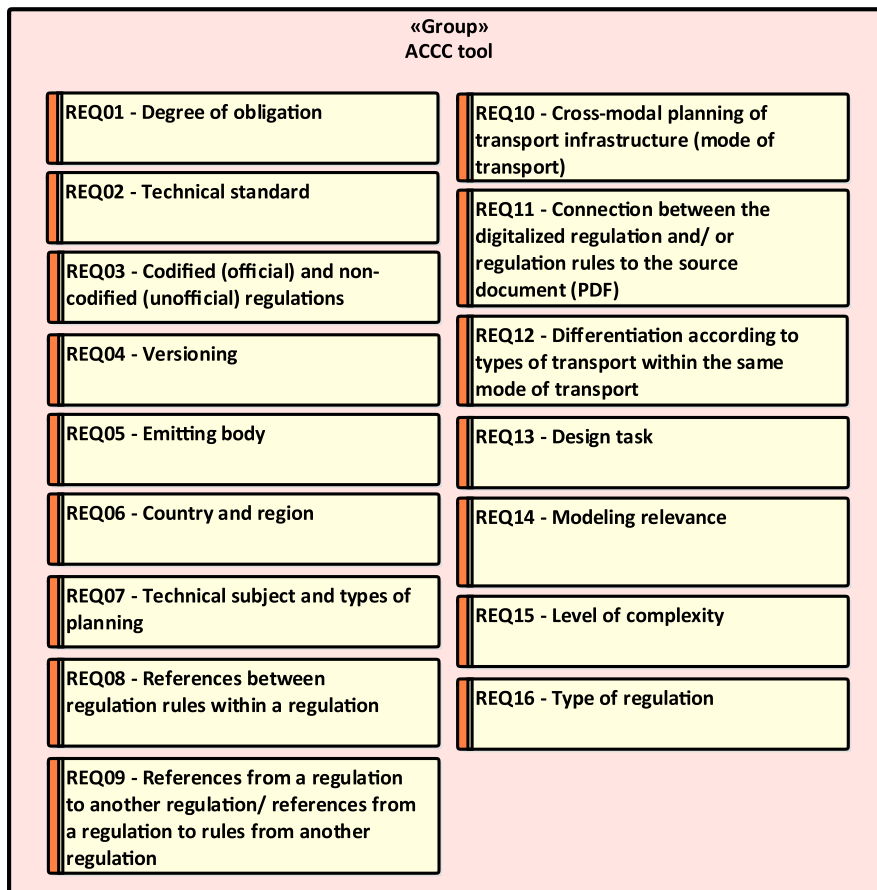


Fig. 4. Requirements to be considered in the development an ACCC tools.

partly based on examples from the German guidelines for railway (*DB AG* guidelines) and road planning (*Road and Transportation Research Association guidelines*, *FGSV* by its German acronym). A summary of the aspects presented in this paper can be found in Fig. 4. Subsequently, the aspects discussed here are used to classify the regulations and rules in a database. The categorization is presented in the Section 4.

REQ01. Degree of obligation (mandatory/ non-mandatory)

The degree of obligation (or bindingness) can be (and will be) defined either for the entire regulation or for the individual regulation rules contained in it. These can have a legal, requirement (commandments/prohibitions), recommendation, information, permissibility or possibility character. It should be mentioned here that the bindingness degree can be expressed either via the technical standard or with the help of modal auxiliary verbs (for example *must*, *must not*, *should*, *should not* etc.), i.e., via the formulation or choice of words. These words usually indicate the degree of obligation.

An example in which the bindingness is indicated by the technical standard, can be taken from the guidelines of the *FGSV*. The *FGSV* guidelines are essentially divided into two categories: R 1 and R 2. The R 1 guidelines have a high obligation degree and correspond to the acknowledged rule of technology. Guidelines in this category specify how a technical circumstance *shall/must* or *should* be dealt with, whereby the explicit use of modal auxiliary verbs in the text is dispensed with. The R 2 guidelines, on the other hand, have a recommendatory character and are assigned to the technology standard state of the science and technology. They indicate how to address these circumstances. It is important to note that each guideline of the *FGSV* can be adopted and introduced by a German supreme state authority via *General Circulars for Road Construction*, which makes a guideline obligatory regardless of its category [20]. An example of the guidelines in which the bindingness of the individual regulation rules is indicated via the modal auxiliary verbs are the guidelines of *DB AG* (e.g., Refs. [15,16]).

Due to their origin, some regulations have innately full degree of obligation. Thus, the laws and legal norms of the legislative and executive power are always binding.

When developing an ACCC tool, one should therefore ask oneself which regulations and/or regulation rules are binding and therefore absolutely must be examined.

REQ02. Technical standard

Regulations and their rules reflect a certain technical standard. The meaning of technical standards is usually classified by levels - according to the so-called *Level theory*. These are, for example, such as the *Three-level theory* according to Refs. [4,5], the *Two-level theory* according to Ref. [57] and the *Unity theory* according to Ref. [19]. In Ref. [41] with reference to Ref. [19], a total of 35 different technical standards are mentioned. This clearly shows that there is no uniform criterion in this respect. The view of the *Three-level theory* has de facto prevailed in practice in Germany [4]. Its structure establishes the following technical standards: acknowledged rule of technology, state of the art and state of the science and technology. The three technical standards are based on the hazard potential and the technical controllability of a circumstance. In this sense, the technology standard acknowledged rule of technology is used for the circumstances with the lowest hazard potential and the highest technical controllability of this hazard potential. For the circumstances with the highest hazard potential and the lowest technical controllability, the technology standard state of the science and technology is used. Accordingly, the technical standard state of the art lies between these two [4]. In other countries, the technical standard may be determined on the basis of other characteristics.

Similar to the degree of obligation, another question that should be asked when developing an ACCC tool is which technology standards must be necessarily checked and which can be left out. This question can apply to an entire regulation as well as to individual regulation rules.

REQ03. Codified (official) and non-codified (unofficial) regulations

It is not always possible to comply with the regulation rules of already established regulations, which in this paper are referred to as codified (official) regulations. In these situations, it could be possible to deviate from them, provided that enough justification is given, and that the safety at all stages of the lifecycle of the infrastructure is guaranteed.

There must be evidence that the situation called for a solution that falls outside of the scope of official regulations and rules and that the proposed solution is adequate from a technical, economical, and safety point of view. This proof is what here is considered as non-codified (unofficial) regulations.

The ACCC tools must be able to identify these situations, so that it would not wrongly recognize them as a non-compliance. In planning, it must also be possible to store and call up the information on the planning circumstances that deviate from certain codified regulation rules.

REQ04. Versioning

It may happen that two identical regulations describe the same technical circumstance or apply to the same technical circumstance, but establish different reference values in the regulation rules. This may be the case if there is a large time difference between two identical regulations, so that the regulation under consideration has evolved or changed over time. Consequently, it must be known with which regulation according to which version (date of issue) the planning of an infrastructure to be checked was carried out. This allows the correct regulation to be selected in the ACCC tool. In the ACCC tool there must also be the possibility to manage the regulations with different versions. In addition, the problem of permanent updating of the ACCC tool should also be mentioned in this context. Easy maintainability should be ensured.

REQ05. Emitting body

Regulations are issued by different standardizing/standards institutions, societies and organizations. It may happen that each institution has different requirements for the same mode of transport for a particular design circumstance and process, procedure, method, etc. Accordingly, when checking an infrastructure project for compliance with the regulation rules, it must be known according to which regulations of which institutions, societies and organizations the planning was made. Therefore, it must be possible to store and manage regulations from different emitting bodies in the ACCC tool.

REQ06. Country and region

Each country usually has its own regulations with their own regulation rules. Consequently, it should be known in which country the infrastructure will be built. The consideration of a planning according to the countries should be possible in an ACCC tool.

Within the country, there are also additional regulations that must be observed only in certain regions, states, etc. This can lead to the fact that in addition to the regulations with a validity for the entire country, regional regulations must also be taken into account. The problem of country and region affiliation is also addressed by the authors in Refs. [30,47,61].

REQ07. Technical subject and types of planning (specific stage of the lifecycle of the project)

Regulations can be oriented to different technical subjects, for example, traffic planning, alignment, traffic safety, economic efficiency, environmental and land protection, specific engineering structures (tunnel, bridges, drainage), etc. At the same time, they can refer to a specific stage of the lifecycle of the project (e. g., planning and design, construction, operation, or maintenance) and to a specific matter (e. g., properties of materials, processes, methods, etc.). When developing an ACCC tool, it is important to keep in mind that a certain regulation can often address several issues at the same time, thus these tools must be able to recognize the specific technical subject, type of planning, and select the appropriate regulations accordingly.

Many of the approaches developed today cover only a part of a technical subject and do not work across different areas (rules from

different regulations for different technical subjects) [59].

REQ08. References between regulation rules within a regulation

Regulations can contain in their regulation rules references to other chapters, modules, etc. within the same document. These references can be seen as relationships between rules, which complement the information contained in the statements, and therefore is necessary that ACCC tools can represent these linkages. This aspect ensures the completeness and comprehensiveness of the checking process.

REQ09. References from a regulation to another regulation/ references from a regulation to rules from another regulation

Regulations can have references to other regulations and/or to individual regulations/modules/chapters etc., of another regulation for the purpose of complementing the description and/or requirements about a technical circumstance (to avoid including unnecessary, too detailed or repeated information in the regulation under consideration). The references can also imply the consideration of additional technical circumstances. It should be noted that these can be *“either dated, undated or general, and at the same time either exclusive or indicative”* [18].

REQ10. Cross-modal planning of transport infrastructure (mode of transport)

Planning can affect several modes of transport at the same time. An example of this is the planning of a railway crossing. A railway crossing is a special crossing situation, since the circumstances related to both road and rail traffic systems must be taken into account at the same time. Accordingly, the regulations of both modes of transport must be taken into account.

Another example is the planning of a tramway. In this case, the rail and road modes of transport must be considered simultaneously too. Consideration should also be made about the fact that the regulations for the different modes of transport can be and are published by the different standardizing/standards institutions, societies and organizations, etc.

REQ11. Connection between the digitalized regulation and/ or regulation rules to the source document (PDF)

Access to the source document (regulation) and its content (regulation rules) must be guaranteed. This direct link between the digital representation of regulations and their source (PDF document) helps to ensure that the correctness of the rule translation can be easily checked. Additionally, it is important to keep in mind that the original document will always have prevalence over the digital translated version.

REQ12. Differentiation according to types of transport within the same mode of transport

In addition to the case where two transport routes of different modes intersect or coincide at some points (REQ10), there is also a possible situation in which several types of transport have to be considered within the same mode of transport. An example of this would be the design of urban roads, where different types of transport such as pedestrian, bicycle, car, stationary and public traffic have to be taken into account. The regulation rules for planning these types of transport are usually described within a single regulation. As an example, the guideline [21] of the FGSV for the planning of urban roads can be mentioned. In this case, the ACCC tool must be able to recognize all the types of transport that need to be considered in each specific case. Additionally, for individual types of transport, there can be further specifications, requirements, and details that need to be regarded. Here, again the references between the different regulations and regulation rules become important. This aspect could thus be supported by the construction of the relationships between regulations and regulation rules (REQ08 and REQ09).

REQ13. Design task

This aspect is directly related to the technical subject and types of planning (REQ07). A planning process or a design task can involve remodeling/expansion, new building, maintenance, etc. of an

infrastructure construction. The differentiation according to the mentioned design tasks can have an influence on the form, consideration and implementation of a regulation rule. Even if, among other things, the same technical circumstances are considered. An example of this is the dimensioning of the superstructure of a railroad line with regard to the protective layers described in the guideline [13] of DB AG, according to which a distinction is made between new construction and improvement/renewal when dimensioning the protective layers.

REQ14. Modeling relevance

Among other things, this aspect is related to the first requirement mentioned, degree of obligation (REQ01). The content of a regulation provides information of various kinds. These represent explicit knowledge, with different character (bindingness). These could also contain instructions, definitions, guidance, remarks, explanations, among others, for the application of a certain rule. The entire content of regulations can be roughly divided into the main body (the actual regulation rules) and additional elements (information that does not have a significant impact on the rules) [17]. Against this background, it should be clarified during the development of an ACCC tool which knowledge of a regulation should necessarily be translated into a machine-readable way. In addition, considerations should be made about the extent to which the knowledge should be made machine-readable and the level of detail required. In this context, it is important to understand which parts of the contents of a regulation cannot be interpreted in a machine-readable way.

The problem of digital translation of regulations has been known for about 40 years, some authors even speak of 50 years. Nevertheless, it has not yet been possible to achieve the necessary level of maturity for a digital regulation checking [30,31,47]. The existing problem was already reported in detail in 1995 in Ref. [52]. Research into the content of regulations in England and Wales has shown that 20% of the rules in regulations can be implemented in a machine-readable way, while 47% require a great deal of adaptation by the user and 33% cannot be implemented at all ([39] with reference to Ref. [54]).

REQ15. Level of complexity

This aspect is defined based on [58], where the author proposes to divide the regulation rules into four categories, depending on their complexity. The least complex rules are those that only require checking data that is explicitly contained in the BIM model. Complexity increases as more complex operations are required (e.g., deriving implicit information from explicitly defined attributes). The applicability of complex rules then depends on the level of detail (Level of Development = LOD) of the BIM model (explicit and derived information), since the higher the LOD, the more detailed the code compliance checking can be.

In the present work, only three complexity level classes are defined, the first one corresponding to the first class (*“Rules that require a single or small number of explicit data”*) from Ref. [58]. Rules of classes 2 and 3 (*“Rules that require simple derived attribute values”* and *“Rules that require extended data structure”*) are combined in the second complexity level class. It was used for simplicity and to ensure that the classification system is exhaustive (ideally, objects belong to only one class of the same rank) [29]. Finally, the rules that belong to class 4 according to Ref. [58] (*“Rules that require a proof of solution”*) are assigned to the third complexity level class. It should be mentioned that the complexity level class 3 automatically includes the complexity level classes 1 and 2.

This aspect should help to identify similar structures in regulations rules, thus supporting the process of translating them into a machine-readable language, by helping to decide on the best suitable methods and tools to be used in each case [58].

REQ16. Type of regulation

Technical regulations can be of various types. They can, for example, convey information (statement), require the execution of an action (instruction), provide guidance (recommendation), etc. [18], with respect to different issues (e.g., terminology, processes, products). These

may be emitted by regulatory and enforcement authorities, as well as other recognized standardizing/standards institutions, societies and organizations. Likewise, there are also different types of regulation documents. For example, laws, legal norms including administrative regulations, directives/rules, technical specifications, instructions for practice, guidelines, etc. This fact should be taken into account in a classification system for regulations, regulation rules and ACCC tool. ACCC tools must be able to recognize the different types of regulations, as this can support the checking process, for instance, by making it easier for the user to search for the required standards through this criterion.

4. Discussion

4.1. Generalities

In order for the ACCC tool to correctly fulfil its purpose, two important prerequisites must be met: (1) To achieve a (semi-)ACCC in BIM, the regulations must be made machine-readable [1]. (2) In addition to the regulations, the infrastructure model must also be digitally available (BIM). This model must be complete and contain all the necessary information.

Basically, two important interacting components must be present for the review process:

- “The building model, a digital representation of the design data, which is subject to audit for compliance
- Normative knowledge, a computable representation of normative provisions stipulated by codes and standards, which is the baseline for the audit” [47]

The structure of machine-readable regulation rules must have a clear ontology and be compatible with the ontology of the BIM objects. A uniform classification is required both for the regulation rules and for the objects of the building [42,47]. This is a topic that is still under research [30].

A uniform classification system of infrastructure objects is an important component for checking a planning for compliance. This should enable the ACCC tools to provide a high verification quality [42]. This opinion is also held by the authors in Ref. [47]. The research carried out on a uniform classification system for infrastructure objects shows that initial considerations exist worldwide at the research level, but these are based on different approaches [3,35,38]. It remains to be considered what the basic structure of a decomposed infrastructure should look like, which frameworks should be used for the basic structure, and in what kind of database this must be implemented.

4.2. Digitalization of explicit knowledge

For completion, formalization, organization, and management of knowledge as well as deletion of repeating planning tasks (repetition of knowledge) there are many techniques that form the so-called *knowledge technologies*. They can be divided into three main domains: Knowledge Engineering (KE), Knowledge Management (KM), and Knowledge-Based Engineering (KBE) [23]. The latter with their Knowledge-based Systems (KBS; also known as the Expert System [23,46]) exhibits the most comprehensive solution and it can be presented as a fully-integrated system with CAD capabilities or as an interface to a CAD tool [23,37] (see Fig. 5).

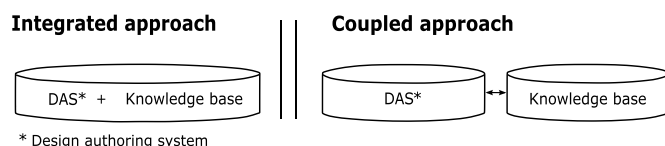


Fig. 5. Types of KBS [37].

With a KBS two objectives are pursued. First, the system should allow the long-term maintenance of engineering knowledge. Second, the system should help the sorted engineering knowledge to find solutions during the planning [45].

Recently, a decoupled KBS for infrastructure construction from the CAD tools was presented in Ref. [37], which formalizes unstructured knowledge and process flows in the railway infrastructure planning through the VPL (BPMN and DMN notations) according to Ref. [28] MOKA (Methodology and tools Oriented to Knowledge-based Engineering Applications). The presented methods to formalize the knowledge should be transparent and correspond to a white-box solution.

A database is needed to machine-readable digitalize, map and manage the engineering knowledge from the traffic road construction (infrastructure objects, regulation rules from the regulations as dependencies between the infrastructure objects, etc.) and to repeatedly have this knowledge available for use in the planning-processes in infrastructure construction. One of the possible solutions could be a graph database. The reason was already mentioned in Section 3.

4.3. Implementation options for the consideration of aspects in the development of an ACCC tool

Once the different requirements that are relevant for the development of an ACCC tool have been described, it must be decided which of them are to be implemented in a database via modeling (modeling nodes and edges), as a category (class structure), or as attributes (class attributes). The subdivision of each aspect in these three levels is shown in Table 1.

The subdivision shown in Table 1 is the result of an own deliberation process. In this process, for example, it was considered that it only makes sense to implement requirements that are either references (REQ08 and REQ09) or connections between objects (REQ11) via relationships.

Table 1

Subdivision of the requirements to be considered for the modeling of regulations and regulation rules in a database according to the classes, class attributes and modeling aspects.

No.	Requirement	Facet/ Class	Class attribute	Modeling (nodes and edges)
REQ01	Degree of obligation	x		
REQ02	Technical standard	x		
REQ03	Codified (official) and non-codified (unofficial) regulations	x		
REQ04	Versioning		x	
REQ05	Emitting body	x		
REQ06	Country and region	x		
REQ07	Technical subject and types of planning	x		
REQ08	References between regulation rules within a regulation			x
REQ09	References from a regulation to another regulation/ references from a regulation to rules from another regulation			x
REQ10	Cross-modal planning of transport infrastructure (mode of transport)	x		
REQ11	Connection between the digitalized regulation and/or regulation rules to the source document (PDF)			x
REQ12	Differentiation according to types of transport within the same mode of transport	x		
REQ13	Design task	x		
REQ14	Modeling relevance	–	–	x
REQ15	Level of complexity	x		
REQ16	Type of regulation	x		

Therefore, these aspects should be implemented via edges. REQ14 concerns the detailing of the machine-readable description of the regulation content and thus the modeling aspect, which is to be implemented in a graph database via the nodes and edges. Thus, this does not depend on the basic class structure of the regulations and regulation rules. Likewise, the regulation-related class attributes are not affected by this aspect. In contrast, REQ04 is considered over the class attribute. Here this aspect would be implemented over an attribute field of the type *date*. A class attribute is meaningful, as long as it does not concern the description of a certain specific characteristic of an individual object, but corresponds to a feature that can be applied to the set of all objects. This applies with the versioning of the regulations, since all objects of the regulations must have a date. Finally, all other requirements are implemented via a class structure of regulations and regulation rules since these aspects are to be combined differently depending on the regulation.

To categorize the regulations and regulation rules, the grouping method by facets has been chosen. With the help of a faceted classification system, several classes can be assigned to an object and thus combined with any properties. This is an advantage in the case where multiple aspects can be used to describe regulations and regulation rules. A facet classification system allows the description of both simple and complex aspects of each subject area to be classified. New facets can be formed and combined with other facets, making it easier to describe the new objects to be recorded [26,29,51]. Within the facets, subclasses can be accommodated, which are related to each other through partitive and generic relationships. In sum, this type of classification system provides a possibility to structure concepts and objects and represent them in a detailed manner, with great flexibility and extensibility. It should be mentioned that the approach presented in this article reflects only the opinion of the authors. There may be other grouping possibilities and alternative solutions to the deliberations presented here. These should be explored in the context of further research.

The next step was to consider which of the aspects could best be implemented at the level of the regulations and which at the level of the regulation rules. The decision was made in particular with regard to the question of whether the individual aspects relate to regulation rules or rather to the entire regulation.

In some cases, it has been useful to consider one or the other aspect at both levels. For example, the requirement technical subject and types of planning (REQ07) of an entire regulation (or of a module from the regulation) can refer to the technical subject for the construction process, but at the same time the regulation rules contained in a certain regulation (or module of the regulation) can also describe other technical circumstances (or subjects). It could also happen that other modules of the same regulations refer to other technical subjects. Against this background, it can be stated that the circumstances of a technical area can either concern the entire regulation, a module of the regulation, or only the individual regulation specifications. A combination is not excluded. Therefore, it should be considered whether some requirements would have to be taken into account at both levels (level of regulation and level of regulation rules). Currently, the REQ07 technical subject and types of planning is only considered at the level of regulations (For this reason, the “x” in Table 2 in the column of regulation rules is in brackets). Another example is the technical standard (REQ02). A certain technical standard could apply to a regulation as well as to the individual regulation rules. Against this background, the question arises as to how detailed one would like to consider this aspect and whether a differentiation should take place at both levels. Currently, this is only taken into account for regulations (which is why in Table 2 the cross in the regulation rules column is in brackets). In the case of REQ13, just as in the case of REQ02 and REQ07, it would also be necessary to consider whether it should be implemented at the level of regulation rules in addition to the level of regulations, thus the corresponding cell in Table 2, under the column regulation rules, is in brackets.

REQ10 cross-modal planning of transport infrastructure (mode of

Table 2

Subdivision of the aspects to be considered for the modeling of regulations and regulation rules in a database into the different levels (regulations and regulation rules).

No.	Requirement	Regulations	Regulation rules
REQ01	Degree of obligation	(x)	x
REQ02	Technical standard	X	(x)
REQ03	Codified (official) and non-codified (unofficial) regulations	X	
REQ04	Versioning	X	
REQ05	Emitting body	X	
REQ06	Country and region	X	
REQ07	Technical subject and types of planning	x	(x)
REQ08	References between regulation rules within a regulation		x
REQ09	References from a regulation to another regulation/references from a regulation to rules from another regulation	x	x
REQ10	Cross-modal planning of transport infrastructure (mode of transport)	x	
REQ11	Connection between the digitalized regulation and/or regulation rules to the source document (PDF)	x	
REQ12	Differentiation according to types of transport within the same mode of transport		x
REQ13	Design task	x	(x)
REQ14	Modeling relevance		x
REQ15	Level of complexity		x
REQ16	Type of regulation	x	

transport) is considered at the level of regulations, whereas the individual types of transport (REQ12 differentiation according to types of transport within the same mode of transport) are considered at the level of regulation rules. This means that the entire regulation will only apply to one specific mode of transport (road, rail, etc). However, their individual regulation rules may affect different types of transport (bike, pedestrians, motorized transport, etc.).

REQ01 degree of obligation has been assigned to the regulations, as this relates to the individual regulations in the narrower sense. However, there are cases in which a certain degree of obligation is valid for the entire regulation regardless of the wording (modal auxiliary verb) (e.g., FGSV guidelines). In such a case, it would have to be considered whether this aspect should also be taken into account at the level of the regulations (which is why in Table 2 the cross in the column regulations is in brackets). The bindingness degree associated to laws or legal norms was implemented via the REQ16 type of regulation at the level of regulations.

The distinction between codified (official) and non-codified (unofficial) regulations (REQ03) is made at the level of the regulations as it relates to the entire regulation (document). The versioning, the country or region to which a regulation belongs, and the emitting bodies are also assigned only to the regulation level since this information is usually found in the general or introductory sections of a regulation and not in the single regulation rules.

References/relationships/connections between the individual regulation rules are implemented in modeling terms via the edges at the level of the regulation rules (REQ08), whereas references/relationships/connections between the individual regulations and rules are implemented at both levels (REQ09). Edges are also used to create links between the modeled regulations and the regulations stored in PDF form in the data base (REQ11). REQ14 (modeling relevance: level of detail of modeling via nodes and edges) and REQ15 (level of complexity) are to be implemented at the level of regulation rules, as these only affect the regulation rules themselves.

4.4. Classification of regulations and regulation rules

Classifying knowledge can be a way of formalizing it. Through classification systems, it is possible to achieve a structured and simple representation of a certain subject, thus facilitating its interpretation. One of the challenges of translating rules into a machine-readable language is that it requires a good understanding of both the technical issues and the structure and composition of the rules and regulations.

In this sense, the classification of regulations and regulation rules should contribute to their enforcement, because it can provide a better understanding of them and a faster identification of the infrastructure objects that are relevant for a certain checking process [2,24,42,43,53]. Additionally, by a uniform classification of regulation rules, one would achieve higher extensibility/modifiability (flexibility) [30].

Therefore, developing a classification system for rules and regulations can be an instrument to optimize the development of ACCC tools, and should bring the following advantages:

- Provide a basic structure of information that can facilitate national and international cooperation by facilitating the comparison and sharing of information [33]
- Easy and fast identification of the regulation rules that would (necessarily) need to be examined in an infrastructure project or in a particular situation [49]
- Regulation rules can have different complexity. This has implications for the translation method of the regulation rules into a machine-readable representation (e.g., additional tools may be required). Identifying similar structures and complexity in the rules and associate this with the translation method can save time in the development of ACCC tools, by allowing the reuse of standardized methods for a given complexity type. Additionally, the technology needs for the development of an ACCC tool can be established [58]
- Easier definition of checking routines/algorithms that speed up the process by determining what to check first and stopping the process if no compliance is found, rather than unnecessarily continuing to check other related higher order circumstances [2].
- Easy identification of the degree of obligation and/or the type of statement to be checked (e.g., exclusive or optional requirement) [18].
- Support the definition of information requirements from the BIM Model by facilitating the identification of the objects to be checked (mapping between the regulations/rules and BIM Model) [25]

The following basic principle has been adopted for the modeling of regulations and regulation rules: regulations and regulation rules are modeled separately as objects of the classification systems established in the data base. This means that both, the regulation and the regulation rules, get their own class structures. The reason for this lies in the discussed requirements, which are to be modeled either at the level of the

regulations, at the level of the regulation rules, or at both levels simultaneously.

4.4.1. Classification of regulations

After listing all the required aspects and assigning them to the level of regulations, the proposed faceted classification system as result for the categorization is shown in the Fig. 6.

The classification system is composed of 8 facets (country and region (REQ06), technical subject and types of planning (REQ07), emitting body (REQ05), design task (REQ13), mode of transport (REQ10), codified/non-codified (REQ03), type of regulation (REQ16) and technical standard (REQ02)), which contain further sub-categorizations underneath.

The DB AG alignment guideline *Ril 800.0110* [10] is taken as an example to illustrate the classification of objects using the proposed system (see Fig. 6, box outlined in blue). The guideline [10] is a codified guideline and applies to the alignment of railways (technical subject: alignment, mode of transport: rail). This guideline considers safety-related, technical and economic aspects throughout Germany and is used both for new and existing railways (speed increases, line improvements). The regulation rules of the guideline *Ril 800.0110* correspond to the technical standard acknowledged rule of technology [10].

The classification system of regulations proposed here is solely the opinion of the authors and is intended to represent only one of the ways of categorizing regulations. Other approaches/solutions cannot be excluded.

4.4.2. Classification of regulation rules

Until now, there has not been much research in the field of classification of regulation rules in infrastructure construction. In a general context, the author of [58] proposes a classification in 4 categories, according to the level of complexity. The proposed categorization should contribute to the development of models for ACCC and even though it was developed using examples from building construction, it is considered to be extensible to other domains, for example, infrastructure.

A more detailed classification of rules in the area of infrastructure construction is presented in Ref. [39]. Here, the authors suggest 12 rule categories (see Table 3) and assign all of them to the third level of complexity defined by Ref. [58].

Having established which of the defined requirements should be considered for the digitalization of regulation rules, the following faceted classification system is proposed (see Fig. 7).

A series of regulation rules were taken from German guidelines in the area of road and rail construction and classified using the proposed system, in order to test its completeness (example carriageway height in Fig. 8).

In addition to the aspects already considered, further considerations were made with regard to the description of a technical circumstance.

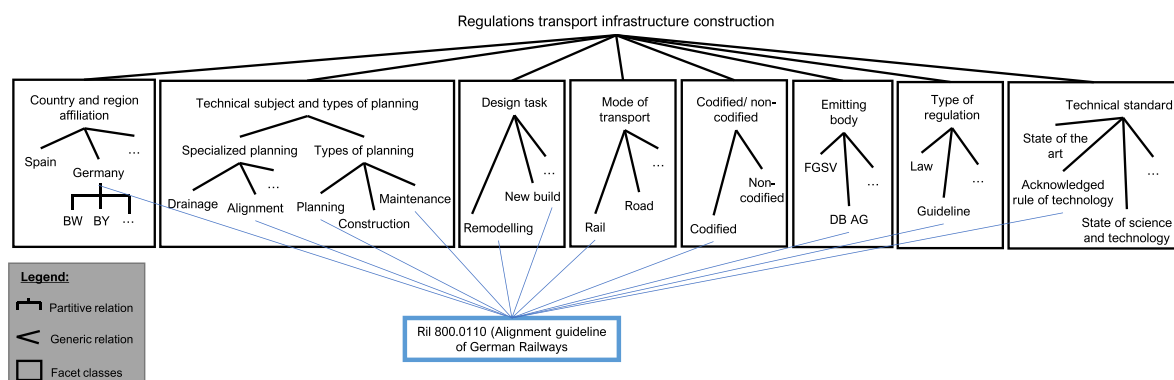


Fig. 6. Suggested classification system for regulations in infrastructure construction.

Table 3

Classification of regulation rules according to Ref. [39].

No.	Rule class	Description
1	Not relevant for modeling	Specifications that have no influence on the modeling of buildings, e.g. procedural processes in communications with the client or specifications for document structures such as explanatory reports.
2	Distance definitions: transverse direction	Definitions (horizontal distances) that can be derived from the cross-section of the building model.
3	Distance definitions: position	Site plan view in the sense of a 3-panel projection: distances between elements.
4	Distance definitions: height	Definitions (vertical distances) that can be derived from the cross-section of the building model.
5	Component definitions	Specifications determining the component or component type to be used.
6	Definitions for alignment elements	Definitions that define parameters for vertical and horizontal alignment elements (straight line, arc, transition arc).
7	Slope definitions	Definitions (slope distances) that can be derived from the cross-section of the building model.
8	Directional definitions: Longitudinal direction	Orientations of components in the layout plan.
9	Clearance tests	Rules for ensuring clearance along railway lines.
10	Other definitions	Rules that do not correspond to the other rule classes but still have an impact on the building model.
11	Construction process	Rules that define temporal components of a construction site.
12	Dimensioning specification	Rules for dimensioning components, such as hydraulic calculations.

This resulted in the following categories:

- geometric information,
- non-geometric information and
- alignment.

In this sense, it was deliberated whether a regulation rule refers to a geometric/non-geometric circumstance or refers to the technical subject alignment.

In infrastructure construction, the checking processes usually focuses on verifying geometric characteristics, because they determine structural features and safety of an infrastructure. This may involve vertical, horizontal or longitudinal geometric dimensions. These quantities are required in the dimensioning of the cross-sections and alignment planning, whereby the technical subject alignment is to be emphasized. The alignment forms the basis of any planning of a traffic route, since it is the reference axis around which the traffic route and other infrastructure facilities are to be built. It represents a special planning step in which

special formulas and physical quantities must be taken into account. This technical subject is therefore given its own category, which is further divided into the sub-categorizations of alignment elements and alignment parameters.

In addition to the geometric circumstances, there are also non-geometric circumstances. These can be, for example, the properties of the construction materials, the type of infrastructure object to be used (e.g., type of sleepers), the sequence of certain objects (for example, construction sequence of layers), the equipment and the number of elements required for a certain infrastructure section or facility, etc. To represent this information, a separate facet has also been added to the system. Other facets were created for the type of transport, the degree of obligation and the level of complexity. In total, the classification system of the regulation rules contains 6 facets (geometric information, non-geometric information, alignment, types of transport (REQ12), degree of obligation (REQ01) and level of complexity (REQ15)) with their sub-categorizations.

Fig. 7 shows an example of the classification of an object (regulation rule) using the proposed system. Fig. 8 illustrates the classified rule, which corresponds to the specifications for determining the height of a railroad line as a function of speed, according to the guideline [10] of DB AG.

This type of technical circumstance specifies a vertical geometric dimension (carriageway height), so it is considered part of the facet geometric information. It does not contain any alignment-related information (alignment parameters or elements) and corresponds to the complexity level of class 2, since several pieces of information are required to verify this regulation rule. As the first input variable, the speed must be known in order to know whether one is in the speed range up to 200 or over 200 km/h. In addition, the type of superstructure (ballast superstructure or slab superstructure) and the information regarding a lightly loaded track (track load up to 10 000 Lt/d and max. v

1 Cross-sections

(1) The carriageway cross-sections for upgraded and new lines should be based on the dimensions given in ...

	Carriageway dimensions [m]	
	$v_e \leq 200$ km/h	$v_e > 200$ km/h
Carriageway height		
with ballast superstructure	0,70	0,76
with slab superstructure	0,50 to 0,70	0,50 to 0,71
in lightly loaded tracks	0,60	

Fig. 8. Regulation rule on permissible carriageway height according to guideline [10] of DB AG (translated by the author).

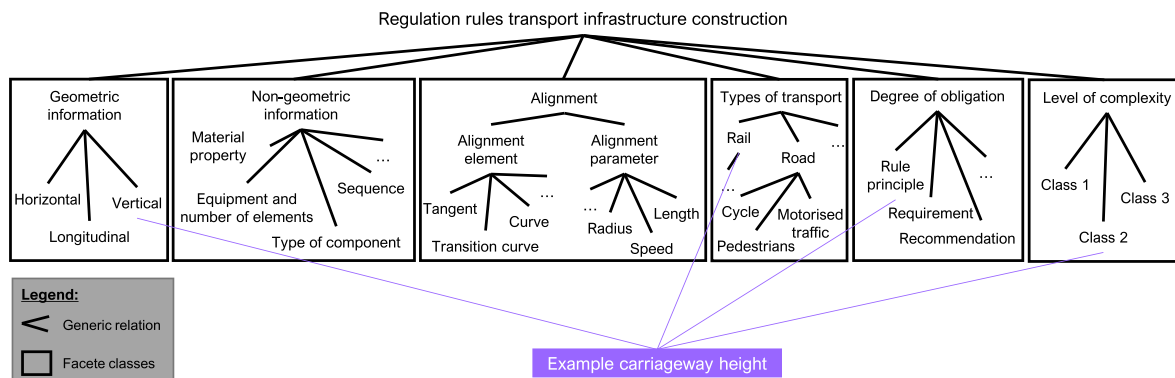


Fig. 7. Proposal for the classification system of regulation rules in infrastructure construction.

= 120 km/h) must be known. Furthermore, it should be noted that the ballast bed thickness for the superstructure with ballast is 0.3 m for speeds up to 200 km/h, and 0.35 m for speeds above 200 km/h. The ballast bed thickness must be increased from 0.35 m to 0.39 m for speeds above 200 km/h. At speeds of over 200 km/h on bridges and in the tunnels, the ballast bed thickness must be increased from 0.35 m to 0.39 m, resulting in a carriageway height of 0.8 m. Only when all this information is known can the carriageway height be derived. The degree of obligation is *rule principle*, since the modal auxiliary verb used in the regulation rule is *should* (see Fig. 8) [10]. According to DB AG, this formulation corresponds to a type of rule that must be adhered to as a fundamental of principle.

4.5. Technical modeling considerations for the digital mapping of regulations and regulation rules

The ACCC process is composed of four parts: (1) rule interpretation (machine-readable interpretation and logical mapping of the rules), (2) building model preparation (preparation of the 3D model for checking), (3) rule execution (the actual checking process), and (4) reporting checking results (presentation of the results) [1,6] (see Fig. 9).

Each step has a significant importance with different associated tasks and functions [50]. Usually the first part, the rules translation/formalization is considered as the most important, and at the same time, the most complicated process [47,50,61]. The regulation rules, which are initially available in a textual form, must remain interpretable by both humans and machines in a suitable form during translation [50,61]. The problem with the texts of the regulations is that they are designed for the reasoning and interpretability of the human brain. These can, for the most part, be easily understood by a human, via logical reasoning among other things, but this type of reasoning represents a great challenge to a machine [42].

The dependencies from the regulations are usually hardcoded in a CAD program, require programming knowledges and do not follow a specific model. Consequently, the background processes remain invisible for the user [1,30]. The implementation is usually carried out in a different way by each software manufacturer. The biggest problem with hard-coded rules is that they cannot be visualized by the users of the software and are therefore not easy to follow or can only be understood with a lot of experience. This makes it difficult to check the plausibility of the results output from a CAD program.

The results of an own conducted survey also showed that the regulation rules are hard-coded, and the rule-checking process is based on the black-box method. However, software vendors are interested in looking for solutions that would allow them to switch from the black-box to the white-box method (see Fig. 10). For storing the regulation rules, usually, a relational database is employed. Due to its limited scalability (capacity to handle a growing amount of work) this type of database can be unsuitable for representing complex planning-related dependencies and relationships.

Generally applicable models for the digitalization and translation of rules to a machine-readable language for their use in ACCC are only

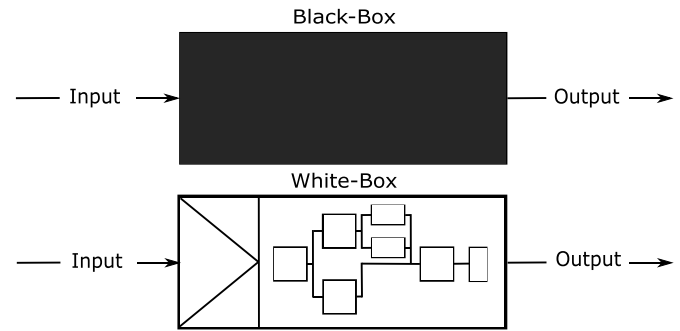


Fig. 10. Schematic representation of the black-box method and white-box methods [1].

available, up to now, for the building construction sector. It is important to note that these existing models are not yet mature enough, but are under continuous development.

The great amount of knowledge required for the different BIM use cases in infrastructure construction could be represented and managed through formalized models in databases. These models must be designed in such a way that their structure and visualization are clear both for the developers and the users. A convenient way to achieve this is through object-oriented programming (ontologies and taxonomies with classes and inheritance hierarchies) because it allows the creation of simple and user-friendly models that do not require high programming knowledges, thus beneficial for civil engineers [23]. This type of programming can be well symbolized through graphs, which provide a clear representation of different types of specific knowledge.

Ontologies are suitable for constructing formalized and explicit representations of concepts ([27] with reference to Ref. [56]) because they depict concepts or classes and the different kinds of relationships that could be established with other entities (instances, logics, properties, constraints) [22,55]. Therefore, they are well suited for representing highly interrelated knowledge and storing it in such a way that is available for further usage in different applications.

With respect to the KBS, a distinction is made in Ref. [23] between so-called RBS (Rule Based System) and FBS (Frame Based System). In an RBS, knowledge is represented in terms of pure IF-THEN rules. However, the authors of [59] state that not all regulation rules can be described with simple IF-THEN-ELSE logics. Under this consideration, it would probably make sense to develop an FBS. In an FBS, there is a so-called frame in which the knowledge is stored in a formalized way. The frame represents a classification system with its instances, taxonomies and ontologies and thus corresponds in this sense to the object-oriented programming paradigm ([23] with reference to Refs. [9, 32,40,44]). Such systems can reflect a significantly more extensive, more complex knowledge compared to pure RBS [23]. Accordingly, it would be appropriate to use FBS.

If a KBS is to be set up in a graph database, it would be useful to define certain modeling principles. In general, the following basic principle applies in a graph database – representing the network-like

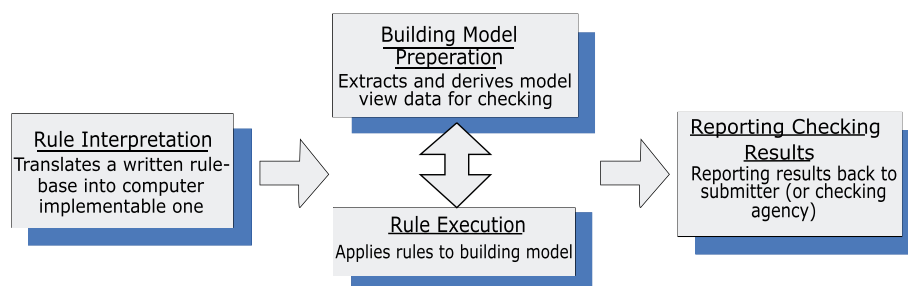


Fig. 9. ACCC process according to Ref. [6].

structures/relationships between the objects (nodes) by means of edges (node-edge construct). This allows an explicit modeling of the connection and dependencies between nodes, contributing to a better visualization and easier traceability. The node-edge constructs could supplement individual models with the so-called modeling attributes. Accordingly, there would be a distinction between modeling edges and modeling attributes in the modeling of the data(sets) (regulations, regulations rules and infrastructure objects). Figuratively, one can imagine this in such a way that the modeling edges are responsible for the coarse model formation and form the basic framework of the models. The modeling attributes of the edges and nodes, on the other hand, are required for fine model building. The modeling attributes would support the development of relatively simpler models, from a visualization and maintainability point of view.

Fig. 11 illustrates a potential ontology for the interaction of regulations – regulation rules – infrastructure objects at a high level of abstraction.

Each individual model shown in Fig. 11 would have its own classification (infrastructure object classification, regulation classification, and regulation rules classification), whereby each classification existing based on a main classification scheme including ontology and taxonomy. This main classification could be used, among others, for further (partial) classification systems for digitally formalizing the explicit knowledge in a given context in infrastructure construction, thus creating a multifunctional data base.

The model of regulation rules would be about forward/backward chaining of conditions representing regulation rules from regulations. The regulation rules represent logical expressions (IF-THEN rules, mathematical rules and hybrids of the two), which are then used by the system as a rule-compliant inference mechanism for finding (matching), for instance, suitable types of infrastructure objects and their compatibility with each other. The basis of this process is the first-order logic, which is supplemented by the modeling attributes in the nodes and edges (see Fig. 12).

Fig. 12 (a) provides an example of the explicit node-edge modeling of the regulation rules. Fig. 12 (b) shows the connection regulation – regulation rules – infrastructure objects using a concrete sleeper as an example. In this example, the verification of the speed and track load represents the so-called parameter check in that the values of the parameters of the regulation rules are compared with the permissible value ranges of the sleeper B70 W-2.4 (connection via PCE edges). Once the sleeper has been found, the appropriate fastening system (W 14 K) and the rail pad (Zw 686a) can be derived via the CCE edges. Modeling by means of the CCE represents in the example in Fig. 12 (b) an alternative to modeling directly via the regulation rules and should be well thought

out. The direct references from one infrastructure object to another can reduce the number of edges that would otherwise come in from the regulation rules.

In principle, three important pieces of information are required to model a planning circumstance: (1) information about the regulation (codified or non-codified), (2) the regulation rules, and (3) the infrastructure object/element. The explicit edge modeling would be done using the three edges:

- connection between the regulation and the regulation rule,
- connection between the regulation and the infrastructure object/element, and
- connection between the regulation rule and the infrastructure object/element.

This way, an infrastructure object/element would be identified simultaneously using two modeling edges (from the regulation rule and the regulation). This means that the edge traversing is done from both sides – from the direction of a regulation and from regulation rules. If, in a particular case, there are no regulation rules for an infrastructure object, or an infrastructure object can be clearly determined without a regulation rule, the infrastructure object would be associated only with a regulation (connection between the regulation and the infrastructure object/element). Assigning regulations to an infrastructure object by means of an edge is also necessary to avoid duplicate modeling of the same infrastructure objects. For example, there are cases where a given infrastructure object appears simultaneously in two or more regulations. Instead, one would model multiple incoming edges from the regulations to one infrastructure object.

5. Conclusion

It is expected, that in the near future the BIM methodology will completely replace the traditional and conventional 2.5D planning methodology. This will result in a significant reduction of the manual effort required to perform planning and design tasks and in an increase in the automation degree for such tasks, which in turn could contribute to a lower probability of error. To achieve this, however, further research and efforts are still required, especially in infrastructure construction. Most of the new technologies, methods, and processes have already been tested and implemented to a greater extent in the building sector, but there is still an enormous need for research in the infrastructure field. The literature review has shown that there are still many gaps in the implementation of the different potential use cases of the BIM methodology for the infrastructure sector. One of these is the ACCC, which still requires a great amount of human effort.

In this article, specifications for the development of an ACCC tool were discussed as well as a consideration regarding the categorization of regulations and regulation rules for their storage in a database. Regarding the specifications presented in this article, future consideration should be made about which of these data should be stored in the IFC files and what would be the best way to do it. For example, the current IFC-Rail already offers a good possibility to represent the semantics and geometry of the infrastructure objects, but not the regulations and their contents (rules). In this area, there is still ongoing research. It is necessary to consider the degree of detail of the classification systems of the regulations and their standards. The categorization should be sufficiently detailed to represent and describe all the required and relevant information, but at the same time general enough to be flexible and applicable in different contexts. This way, different application possibilities are ensured.

To achieve digital representations of engineering knowledge that is related to regulation and regulation rules, technical and modeling possibilities must be considered. In the framework of this article, these considerations were made by means of the possibilities offered by graph database and object-oriented programming. For this, an ontology in the

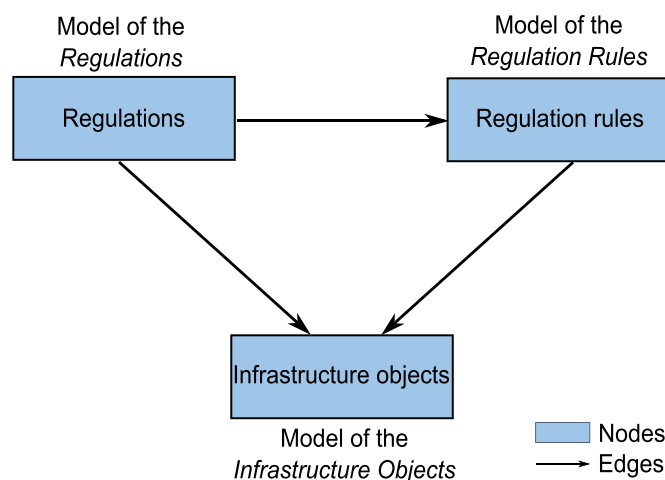


Fig. 11. Suggested ontology for regulations – regulation rules – infrastructure objects in a graph database.

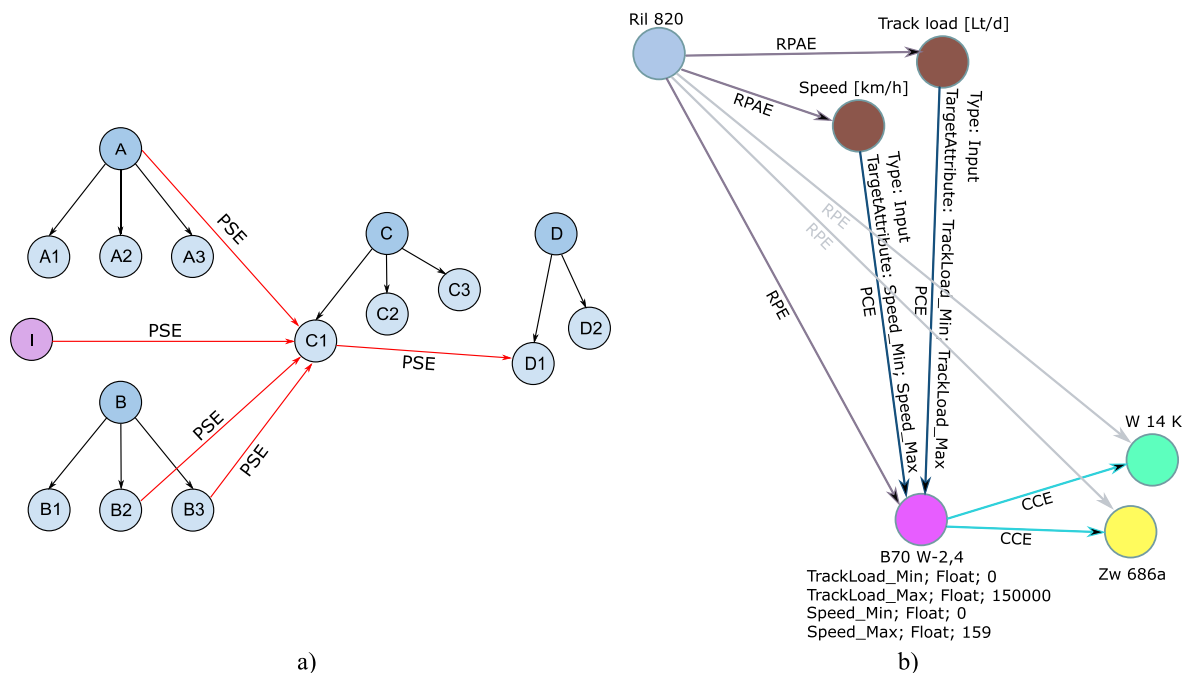


Fig. 12. Modeling of regulation rules and dependencies between regulation rules and infrastructure objects. (a) Example for the modeling the dependencies of the regulation rules via nodes and edges. The dependencies between the individual regulation rules are modeled in the example using PSE edges. (b) Example for the determination of the infrastructure objects using the example of the concrete sleeper B70 W-2.4 and the associated fastening system (W 14 K) and rail pad (Zw 686a) under boundary conditions speed and track load according to DB AG guideline Ril 820 [12] for the dimensioning of the superstructure with ballast.

form of an explicit node-edge construct was proposed. This ontology should support modeling the regulations – regulation rules – infrastructure objects interaction. This should facilitate representing the regulation-dependent interaction of the infrastructure objects/elements needed to plan a traffic route, to allow the development of the planning process under consideration of regulatory constraints for a certain design task. Further work should include the development of modeling and software technical details for the machine-readable representation of the engineering knowledge, which could be carried out with the help of programmers.

CRediT author statement

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Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

No data was used for the research described in the article.

References

- [1] A. Borrmann, et al., *Building Information Modeling Technologische Grundlagen und industrielle Praxis*, VDI-Buch, Springer Vieweg, Springer Fachmedien Wiesbaden, 2021. ISBN: 978-3-658-33361-4.
- [2] A. Yurchyshyna, A. Zarli, An ontology-based approach for formalisation and semantic organisation of conformance requirements in construction, *Autom. Construct.* 18 (8) (2009) 1084–1098, <https://doi.org/10.1016/j.autcon.2009.07.008>.
- [3] buildingSMART Germany, *BIM-Klassen der Verkehrswege, Vorstandardisierungsarbeit der buildingSMART-Fachgruppe BIM-Verkehrsweg*, 2020. ISBN: 978-3-948742-04-1.
- [4] Bundesministerium der Justiz, Bundesanzeiger – Bekanntmachung des Handbuchs der Rechtsförmlichkeit, Bmj.de, 2008. available at: https://www.bmj.de/DE/The men/RechtssetzungBuerokratieabbau/HDR/HDR_node.html (Accessed 05 September 2022).
- [5] Bundesverfassungsgericht, Beschluss vom 8. August 1978 – 2 BvL 8/77 – Kalkar, BVerGE, vol. 49, 1979 available at: <https://openjur.de/u/166332.html> (Accessed 5 September 2022).
- [6] C. Eastman, et al., Automatic rule-based checking of building designs, *Autom. Construct.* 18 (8) (2009) 1011–1033, <https://doi.org/10.1016/j.autcon.2009.07.002>.
- [7] C. Preidel, A. Borrmann, Automated Code Compliance Checking Based on a Visual Language and Building Information Modeling, in: *International Symposium on Automation and Robotics in Construction and Mining (ISARC 2015)* at Oulu, 2015, <https://doi.org/10.13140/RG.2.1.1542.2805>.
- [8] C.Y. Lee, et al., Automated rule-based checking for the validation of accessibility and visibility of a building information model, in: *Computing in Civil Engineering 2015*, American Society of Civil Engineers, 2015, <https://doi.org/10.1061/9780784479247.071>.
- [9] D. Auty, in: NASA (Ed.), *Object Oriented Programming Systems and Frame Representations, an Investigation of Programming Paradigms*, 1988. Technical Report, available at: <https://ntrs.nasa.gov/citations/19900010442> (Accessed 5 September 2022).
- [10] DB Netz AG., *Richtlinie Ril 800.0110 – Streckenquerschnitte auf Erdkörpern* (Guideline Ril 800.00110 – Cross sections at earthwork), 2018. item number: Z800.0110.
- [11] DB Netz AG., *Richtlinie Ril 800.0130 – Streckenquerschnitte auf Erdkörpern* (Guideline Ril 800.00130 – Cross sections at earthwork), 2018. item number: Z800.0130.
- [12] DB Netz AG., *Richtlinie Ril 820 – Grundlagen des Oberbaus* (Guideline Ril 820 Guideline 820 – Basics of superstructure), 2019. item number: Z82001.
- [13] DB Netz AG., *Ril 836 – Erdbauwerke und sonstige geotechnische Bauwerke planen, bauen und instandhalten* (Guideline 836 – Design, built and operate earthworks and other geotechnical constructions), 2019. item number: Z83601.
- [14] DB Netz AG., *Richtlinie Ril 800.0120 – Auswahl der Weichen, Kreuzungen und Hemmschuhauswurfvorrichtungen* (Guideline Ril 800.0120 – Select of switches, slip switches and brake shoes), 2018. item number: Z800.0120.
- [15] DB Netz AG., *Richtlinie Ril 804 – Eisenbahnbrücken (und sonstige Ingenieurbauwerke) planen, bauen und instand halten* (Guideline Ril 804 – Design, built and operate bridges and other engineering works), 2018. item number: D804.XXXX.
- [16] A.G. DB Netz, *Richtlinie Ril 853 – Eisenbahntunnel planen, bauen und instand halten* (Guideline Ril 853 – Design, built and operate railway tunnels), 2018. item number: D853.

- [17] DIN Deutsches Institut für Normung e. V., DIN 820-2, Standardization – Part 2: Presentation of Documents, ISO/IEC Directives – Part 2: 2018, Modified, German and English Version CEN/CENELEC Internal Regulations – Part 3: 2019, 2020. ICS: 01.120.
- [18] DIN Deutsches Institut für Normung e. V., DIN EN 45020, Standardization and Related Activities – General Vocabulary, ISO/IEC Guide 2: 2004, Trilingual version EN 45020: 2006, 2007. ICS: 01.040.01.
- [19] F. Nicklisch, Funktion und Bedeutung technischer Standards in der Rechtsordnung, 1983.
- [20] Forschungsgesellschaft für Straßen- und Verkehrswesen (Road and Transportation Research Association), Grundlagen für das Erstellen von Technischen Regelwerken und Wissensdokumenten für das Straßen- und Verkehrswesen (Basics principles for the creation of technical regulations and knowledge documents for road and traffic engineering), FGSV-Verlag, Köln, 2013. ISBN: 978-3-86446-025-8.
- [21] Forschungsgesellschaft für Straßen- und Verkehrswesen (Road and Transportation Research Association), Richtlinien für die Anlage von Stadtstraßen – RAS (R 1) (Directives for the Design of Urban Roads), FGSV 200, FGSV-Verlag, Köln, 2006. ISBN: 978-3-939715-21-4.
- [22] G. Chen, Y. Luo, A BIM and ontology-based intelligent application framework, in: 2016 IEEE Advanced Information Management, Communicates, Electronic and Automation Control Conference (IMCEC), IEEE, 2016, pp. 494–497, <https://doi.org/10.1109/IMCEC.2016.7867261>.
- [23] G. La Rocca, Knowledge based engineering: between AI and CAD. Review of a language based technology to support engineering design, Adv. Eng. Inf. 26 (2) (2012) 159–179, <https://doi.org/10.1016/j.aei.2012.02.002>.
- [24] G.T. Lau, K.H. Law, An information infrastructure for comparing accessibility regulations and related information from multiple sources, in: Proceedings of 10th International Conference on Computing in Civil and Building Engineering, 2004 available at: https://www.researchgate.net/publication/250443580_An_Information_Infrastructure_for_Comparing_Accessibility_Regulations_and_Related_Information_from_Multiple_Sources (Accessed 05 September 2022).
- [25] H. Luo, P. Gong, A BIM-based code compliance checking process of deep foundation construction plans, J. Intell. Rob. Syst. 79 (3–4) (2015) 549–576, <https://doi.org/10.1007/s10846-014-0120-z>.
- [26] ISO - International Organization for Standardization, ISO 12006-2, Building Construction—Organization of Information about Construction Works—Part 2: Framework for Classification, ICS, 2015, 91.010.01.
- [27] ISO - International Organization for Standardization, ISO/IEC TR 11179-2, Information Technology – Metadata Registries (MDR) – Part 2: Classification, ICS, 2019, 35.040.50.
- [28] ISO - International Organization for Standardization/IEC – International Electrotechnical Commission, ISO/IEC 19510, Information Technology – Object Management Group Business Process Model and Notation, ICS, 2013, 35.020.
- [29] J. Cann, Principles of Classification, NBS Services UK, 1997 available at: https://ici.s.org/wp-content/uploads/2017/10/1997-06_Paper.Principles_of_Classification.pdf (Accessed 05 September 2022).
- [30] J. Dimyadi, R. Amor, Automated building code compliance checking – where is it at?, in: 19th International CIB World Building Congress, 2013 available at: http://www.researchgate.net/publication/303206027_Automated_Building_Code_Compliance_Checking_Where_is_it_at (Accessed 31 August 2022).
- [31] J.H. Garrett, et al., Delivering the infrastructure for digital building regulations, J. Comput. Civ. Eng. 28 (2) (2014), [https://doi.org/10.1061/\(ASCE\)CP.1943-5487.0000369](https://doi.org/10.1061/(ASCE)CP.1943-5487.0000369).
- [32] J. Rumbaugh, et al., Object-oriented Modeling and Design, Prentice-Hall, Englewood Cliff, New Jersey, 1991. ISBN: 0136298419.
- [33] K. Afsari, M. Eastman, A comparison of construction classification systems used for classifying building product models, in: 52nd ASC Annual International Conference Proceedings, 2016, <https://doi.org/10.13140/RG.2.2.20388.27529>.
- [34] L. Khemlani, Analysis, Research, and Review of AEC Technology, 2004.
- [35] M.B. Seo, D. Lee, Development of railway infrastructure BIM prototype libraries, Appl. Sci. (Basel, Switzerland) 10 (22) (2020) 8118, <https://doi.org/10.3390/app10228118>.
- [36] M.F. Porto, et al., Automatic analysis of standards in rail projects, in: The 19th World Multi-Conference on Systemics, Cybernetics and Informatics: WMSCI 2015, 2015 available at: <http://www.iijsci.org/Journal/pdv/sci/pdfs/MA969EM15.pdf> (Accessed 31 August 2022).
- [37] M. Häußler, A. Borrmann, Knowledge-based engineering in the context of railway design by integrating BIM, BPMN, DMN and the methodology for knowledge-based engineering applications (MOKA), J. Inf. Technol. Construct. 26 (12) (2021) 193–226, <https://doi.org/10.36680/j.itcon.2021.012>.
- [38] M. Häußler, A. Borrmann, Model-based quality assurance in railway infrastructure planning, Autom. Construct. 109 (2020), 102971, <https://doi.org/10.1016/j.autcon.2019.102971>, 102971.
- [39] M. Häußler, et al., Code compliance checking of railway designs by integrating BIM, BPMN and DMN, Autom. Construct. 121 (2021), 103427, <https://doi.org/10.1016/j.autcon.2020.103427>, 103427.
- [40] M. Negnevitsky, Artificial Intelligence – A Guide to Intelligent System, Addison-Wesley, Harlow, England, 2005. ISBN: 1408225743.
- [41] M. Seibel, Abgrenzung der “allgemein anerkannten Regeln der Technik“ vom “Stand der Technik“, Neue Jurist. Wochenschr. (NJW) 41 (2013).
- [42] N. Nawari, The challenge of computerizing building codes in a BIM environment, in: Computing in Civil Engineering (2012), American Society of Civil Engineers, 2012, <https://doi.org/10.1061/9780784412343.0036>.
- [43] N.O. Nawari, SmartCodes and BIM, in: Proceedings of the Structures Congress 2013, 2013, <https://doi.org/10.1061/9780784412848.082>.
- [44] O. Lassila, Frames or objects, or both?, in: 8th AAAI National Conference on Artificial Intelligence, Object-Oriented Programming in AI, 1990 available at: <http://www.lassila.org/publications/1990/frames-or-objects.pdf> (Accessed 05 September 2022).
- [45] P. Bermell-García, I.-S. Fan, A KBE System for the design of wind tunnel models using reusable knowledge components, in: Ponencia presentada en el VI Congreso Internacional de Ingeniería de Proyectos celebrada en Barcelona en el año 2002, 2002 available at: https://www.researchgate.net/publication/264879537_A_KBE_SYSTEM_FOR_THE_DESIGN_OF_WIND_TUNNEL_MODELS_USING_REUSABLE_KNOWLEDGE_COMPONENTS (Accessed 05 September 2022).
- [46] P.K. Tripathi, A review on knowledge-based expert system: concept and architecture, in: Artificial Intelligence Techniques - Novel Approaches & Practical Applications, 2011 available at: https://www.ijcaonline.org/ait/number4/SP_E226T.pdf (Accessed 05 September 2022).
- [47] R. Amor, J. Dimyadi, The promise of automated compliance checking, Dev. Build Environ. 5 (100039) (2021), 100039, <https://doi.org/10.1016/j.dibe.2020.100039>.
- [48] R. Angles, C. Gutierrez, Survey of graph database models, ACM Comput. Surv. 40 (1) (2008) 1–39, <https://doi.org/10.1145/1322432.1322433>.
- [49] R. Hussain, et al., Safety regulation classification system to support BIM based safety management, in: 34th International Symposium on Automation and Robotics in Construction (ISARC 2017), 2017, pp. 30–36, <https://doi.org/10.22260/ISARC2017/0004>.
- [50] S.A. Ismail, et al., A review on BIM-based automated code compliance checking system, in: 2017 International Conference on Research and Innovation in Information Systems (ICRIIS), IEEE, 2017, pp. 1–6, <https://doi.org/10.1109/ICRIIS.2017.8002486>.
- [51] S. Batley, Classification in Theory and Practice, Chandos Publishing, 2005, ISBN 1843340836.
- [52] S.J. Fennes, et al., Computer representations of design standards and building codes, U.S. Dep. Civ. Environ. Eng. Int. J. Construct. Inf. Technol. 3 (1995) 13–34, available at: https://www.researchgate.net/publication/30869803_Computer_representations_of_design_standards_and_building_codes_US_perspective (Accessed 05 September 2022).
- [53] S.M. Ilala, H.M. Günaydin, Computer representation of building codes for automated compliance checking, Autom. Construct. 82 (2017) 43–58, <https://doi.org/10.1016/j.autcon.2017.06.018>.
- [54] S. Malsane, et al., Development of an object model for automated compliance checking, Autom. Construct. 49 (2015) 51–58, <https://doi.org/10.1016/j.autcon.2014.10.004>.
- [55] S. Wu, et al., Natural-language-based intelligent retrieval engine for BIM object, Comput. Ind. 108 (2019) 73–88, <https://doi.org/10.1016/j.compind.2019.02.016>.
- [56] T.R. Gruber, Toward principles for the design of ontologies used for knowledge sharing, Int. J. Hum. Comput. Stud. 43 (5–6) (1995) 907–928, <https://doi.org/10.1006/ijhc.1995.1081>.
- [57] U. Battis, C. Gussy, Technische Normen im Baurecht, Rdnr. 288 ff, 1988. ISBN: 3804112404.
- [58] W. Solihin, C. Eastman, Classification of rules for automated BIM rule checking development, Autom. Construct. 53 (2015) 69–82, <https://doi.org/10.1016/j.autcon.2015.03.003>.
- [59] X. Tan, et al., Automated code compliance checking for building envelope design, J. Comput. Civ. Eng. 24 (2) (2010), [https://doi.org/10.1061/\(ASCE\)0887-3801\(2010\)24:2\(203\)](https://doi.org/10.1061/(ASCE)0887-3801(2010)24:2(203)).
- [60] X. Xu, H. Cai, Semantic approach to compliance checking of underground utilities, Autom. Construct. 109 (2020), 103006, <https://doi.org/10.1016/j.autcon.2019.103006>, 103006.
- [61] Y.-C. Lee, et al., A comparative analysis of five rule-based model checking platforms, in: Construction Research Congress 2020, 2020, <https://doi.org/10.1061/9780784482865.119>.